NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 6

TOPIC : WRITE A PROGRAM OF DATA ABSTRACTION, EXCEPTION HANDLING AND SHIFT OPERATORS IN C++

COURSE CODE : CSE 0612-2122

COURSE TITLE : OBJECT ORIENTED PROGRAMMING SESSIONAL.

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Write a program of Data Abstraction, Exception handling & Shift operators in C++:

Data Abstraction:

Code:

```
#include <iostream>
using namespace std;

class Rectangle {
private:
    double length;
    double width;

public:
    void setDimensions(double l, double w) {
        length = l;
        width = w;
    }

    double getArea() {
        return length * width;
    }
};

int main() {
    Rectangle rect;
    double length, width;

    cout << "Enter the length of the rectangle: ";
    cin >> length;
    cout << "Enter the width of the rectangle: ";
    cin >> width;

    rect.setDimensions(length, width);

    cout << "Area of the rectangle: " << rect.getArea() << endl;
    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the length of the rectangle: 4
Enter the width of the rectangle: 6
Area of the rectangle: 24
PS C:\Users\Lenovo\Documents\c++>
```

Exception handling:

Code:

```
#include <iostream>
using namespace std;

double divide(double numerator, double denominator) {
    if (denominator == 0) {
        throw runtime_error("Division by zero error");
    }
    return numerator / denominator;
}

int main() {
    double num, den;
    cout << "Enter the numerator: ";
    cin >> num;
    cout << "Enter the denominator: ";
    cin >> den;

    try {
        double result = divide(num, den);
        cout << "Result: " << result << endl;
    } catch (const runtime_error &e) {
        cout << "Exception caught: " << e.what() << endl;
    }

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the numerator: 5
Enter the denominator: 0
Exception caught: Division by zero error
PS C:\Users\Lenovo\Documents\c++>
```

Shift operators:

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num = 20; // Binary: 10100

    int leftShift = num << 2;
    cout << "Left Shift: " << num << " << 2 = " << leftShift <<
endl;

    int rightShift = num >> 2;
```

```
    cout << "Right Shift: " << num << " >> 2 = " << rightShift <<
endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Left Shift: 20 << 2 = 80
Right Shift: 20 >> 2 = 5
PS C:\Users\Lenovo\Documents\c++>
```